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CASE STUDY REPORT

TOPIC: TELECOM CUSTOMER CHURN – PREDICTION AND SEGMENTATION USING MACHINE LEARNING

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PROFESSIONAL ELECTIVE -I
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TELECOM CUSTOMER CHURN – PREDICTION AND SEGMENTATION USING MACHINE LEARNING

ABSTRACT

Customer churn is one of the major challenges faced by the telecommunications industry. Customer churn refers to the situation in which customers discontinue their services or switch to another service provider. High customer churn directly affects company revenue, customer relationships and business growth. Therefore, predicting customer churn and understanding customer behaviour are important for developing effective customer retention strategies.

This project, titled **"Telecom Customer Churn – Prediction and Segmentation Using Machine Learning,"** focuses on developing a machine learning-based approach to predict customer churn and identify different customer groups based on their characteristics and service usage patterns.

The IBM Telco Customer Churn dataset, containing approximately 7,000 customer records, is used for analysis. Initially, data preprocessing and exploratory data analysis are performed to identify missing values, inconsistencies, customer characteristics and relationships between different attributes.

Three classification algorithms, namely Logistic Regression, Decision Tree and K-Nearest Neighbors (K-NN), are implemented to predict whether a customer is likely to leave the telecom service. Their performance is evaluated using training and testing datasets with metrics such as accuracy, precision, recall, F1-score and confusion matrix.

Additionally, Linear Regression and Polynomial Regression are applied to estimate monthly customer charges. Customer segmentation is performed using K-Means and Hierarchical Clustering techniques to identify groups of customers with similar characteristics.

Based on the prediction and segmentation results, appropriate customer retention strategies are recommended for different customer groups. The proposed approach helps telecom companies identify customers who are at risk of leaving, understand customer behaviour and improve customer satisfaction.

Keywords: Customer Churn, Machine Learning, Logistic Regression, Decision Tree, K-NN, Linear Regression, K-Means Clustering, Hierarchical Clustering, Customer Segmentation.

CHAPTER 1: INTRODUCTION

1.1 Introduction

The telecommunications industry is experiencing rapid growth due to increasing demand for internet connectivity, mobile communication and digital services. Customers have multiple choices when selecting telecom service providers. As a result, competition among telecom companies has increased significantly.

Customer churn is an important business problem in this industry. It occurs when a customer terminates an existing service subscription and moves to another provider or stops using the service.

Acquiring new customers generally requires additional marketing and operational expenses. Therefore, retaining existing customers is an important strategy for maintaining business profitability.

Machine learning techniques can analyse historical customer data and identify patterns associated with customer churn. By applying predictive algorithms, telecom companies can identify customers who have a higher probability of leaving their services.

This project combines classification, regression and clustering techniques to analyse customer behaviour.

1.2 Background of the Study

Telecom companies collect large amounts of customer information, including:

- Customer demographic details.
- Internet and phone service subscriptions.
- Contract information.
- Monthly and total charges.
- Payment methods.
- Customer tenure.
- Customer churn status.

These data attributes can be analysed using machine learning algorithms to understand customer behaviour.

The proposed project uses three major analytical approaches:

1. **Classification:** Predicting whether a customer will churn.
2. **Regression:** Estimating monthly customer charges.
3. **Clustering:** Grouping customers based on similar characteristics.

Combining these approaches provides a comprehensive understanding of telecom customer behaviour.

1.3 Problem Statement

Telecommunication companies face challenges in identifying customers who are likely to discontinue their services. Traditional customer management systems often depend on manual analysis and historical reports, which may not identify potential churn customers at an early stage.

There is a need for an intelligent analytical system that can:

- Predict customer churn using machine learning algorithms.
- Compare the performance of different classification models.
- Estimate monthly customer charges.
- Identify customer groups using clustering algorithms.
- Recommend suitable retention strategies for different customer segments.

This project addresses these challenges using machine learning techniques applied to the IBM Telco Customer Churn dataset.

1.4 Objectives

The primary objectives of this project are:

1. To analyse telecom customer information using exploratory data analysis.
2. To preprocess customer data and handle missing or inconsistent values.
3. To develop churn prediction models using Logistic Regression, Decision Tree and K-NN.
4. To compare the performance of classification algorithms using train-test evaluation.
5. To estimate monthly charges using Linear Regression and Polynomial Regression.
6. To segment customers using K-Means and Hierarchical Clustering.
7. To identify important customer characteristics associated with churn.
8. To recommend customer retention strategies based on customer segments.

1.5 Scope of the Project

The project focuses on machine learning-based customer analysis using publicly available telecom customer data.

The scope includes:

- Customer data preprocessing.
- Exploratory data analysis.
- Classification model development.
- Regression analysis.
- Customer segmentation.
- Model performance comparison.
- Business recommendations.

The project can be extended in the future by developing a web-based dashboard for real-time customer churn monitoring.

1.6 Sustainable Development Goals

SDG 8 – Decent Work and Economic Growth

The project supports business sustainability by helping telecom companies improve customer retention, operational efficiency and revenue stability.

SDG 9 – Industry, Innovation and Infrastructure

The project promotes the use of machine learning and data analytics in telecommunications infrastructure and digital business operations.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Machine learning has become an important technology for analysing customer behaviour and predicting customer churn. Researchers have explored several classification, regression and clustering techniques to improve customer retention.

2.2 Customer Churn Prediction

Customer churn prediction involves identifying customers who are likely to discontinue a service. Historical customer information is analysed to discover patterns associated with churn.

Common factors influencing churn include:

- Contract type.
- Monthly charges.
- Customer tenure.
- Internet service type.
- Payment method.
- Technical support availability.

Classification algorithms are commonly used to predict churn.

2.3 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. It estimates the probability of a customer belonging to a particular class.

It uses the sigmoid function:

$$P(Y=1) = 1 / (1 + e^{(-z)})$$

where:

$z = b0 + b1x1 + b2x2 + ... + bnxn$

The output probability is used to classify customers into churn and non-churn categories.

2.4 Decision Tree

Decision Tree is a supervised learning algorithm that uses a tree-like structure to make predictions.

It divides customer records into smaller groups based on feature values.

Important decision tree concepts include:

- Root node.
- Internal nodes.
- Decision rules.
- Leaf nodes.
- Gini impurity.
- Information gain.

Decision trees are useful because their predictions can be interpreted through decision rules.

2.5 K-Nearest Neighbors

K-Nearest Neighbors is a supervised classification algorithm that classifies a customer based on the classes of its nearest training observations.

Euclidean distance is commonly used:

$d(x,y) = sqrt(sum((xi - yi)^2))$

The value of K determines the number of neighbouring observations considered for classification.

2.6 Regression Analysis

Regression algorithms are used to estimate continuous numerical values.

Linear Regression estimates a relationship between dependent and independent variables.

Polynomial Regression extends linear regression by incorporating polynomial features to capture nonlinear relationships.

2.7 Customer Segmentation

Customer segmentation groups customers based on similar characteristics.

K-Means clustering divides observations into K clusters by minimizing the within-cluster sum of squared distances.

Hierarchical Clustering creates a hierarchy of customer groups using either agglomerative or divisive approaches.

2.8 Research Gap

Although individual machine learning algorithms are widely used for customer churn prediction, analysing churn prediction, charge estimation and customer segmentation together can provide additional business insights.

This project integrates these three analytical tasks into a single workflow to support data-driven customer retention decisions.

CHAPTER 3: DATASET DESCRIPTION

3.1 Dataset Overview

The IBM Telco Customer Churn dataset is a publicly available dataset used for analysing customer behaviour in the telecommunications industry.

Dataset characteristics:

Attribute	Description
Dataset name	IBM Telco Customer Churn
Total records	Approximately 7,043
Total columns	21
Domain	Telecommunications
Data type	Structured customer data
Target variable	Churn
File format	CSV

3.2 Dataset Attributes

Important attributes include:

Attribute	Description
customerID	Unique customer identifier
gender	Customer gender
SeniorCitizen	Senior citizen indicator
Partner	Partner status
Dependents	Dependent status

tenure	Number of months with company
PhoneService	Phone service subscription
InternetService	Internet service type
OnlineSecurity	Online security subscription
TechSupport	Technical support subscription
Contract	Contract type
PaperlessBilling	Paperless billing status
PaymentMethod	Payment method
MonthlyCharges	Monthly service charges
TotalCharges	Total charges
Churn	Customer churn status

3.3 Target Variable

The target variable is Churn.

It contains two categories:

- Yes – Customer has discontinued the service.
- No – Customer continues using the service.

The classification algorithms use this variable to predict customer churn.

3.4 Independent Variables

The independent variables include customer tenure, contract type, monthly charges, internet services, payment methods and other customer characteristics.

The customerID column is excluded from predictive modelling because it is an identifier rather than a meaningful behavioural feature.

CHAPTER 4: PROPOSED METHODOLOGY

4.1 System Architecture

The proposed system consists of the following stages:

1. Dataset collection.
2. Data preprocessing.
3. Exploratory data analysis.

4. Feature selection and transformation.
5. Classification model development.
6. Regression model development.
7. Customer clustering.
8. Performance evaluation.
9. Customer retention recommendations.

4.2 Data Preprocessing

Data preprocessing improves the quality of the dataset before applying machine learning algorithms.

The following preprocessing activities are performed:

Missing Value Handling

Missing values are identified using Pandas. Missing numerical values can be handled using median imputation, while categorical values can be handled using the most frequent category.

Duplicate Removal

Duplicate customer records are identified and removed where appropriate.

Data Type Conversion

The TotalCharges column may contain blank strings. These values are converted into numerical values using appropriate data cleaning techniques.

Categorical Encoding

Categorical variables are transformed into numerical representations.

One-hot encoding is applied to nominal categorical variables.

Feature Scaling

StandardScaler is applied to numerical features where required, especially for Logistic Regression, K-NN and clustering.

4.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the structure and characteristics of the dataset.

The following analyses are conducted:

- Distribution of churn and non-churn customers.
- Customer tenure distribution.
- Monthly charges distribution.

- Churn by contract type.
- Churn by payment method.
- Churn by internet service.
- Relationship between tenure and monthly charges.

Matplotlib and Seaborn are used for visualization.

4.4 Classification Models

Three classification algorithms are implemented.

4.4.1 Logistic Regression

Logistic Regression estimates the probability of churn based on customer attributes.

Steps:

1. Load the preprocessed dataset.
2. Separate input features and target variable.
3. Split data into training and testing sets.
4. Apply feature scaling and encoding through a preprocessing pipeline.
5. Train the Logistic Regression model.
6. Predict churn on the testing dataset.
7. Evaluate the model.

4.4.2 Decision Tree Classifier

Decision Tree identifies customer characteristics that influence churn.

Steps:

1. Initialize the Decision Tree classifier.
2. Train the model using training data.
3. Generate predictions.
4. Calculate evaluation metrics.
5. Analyse decision rules and feature importance.

4.4.3 K-Nearest Neighbors

K-NN classifies customers based on their similarity to neighbouring observations.

Steps:

1. Normalize numerical features.

2. Select an appropriate K value.
3. Train the K-NN classifier.
4. Predict customer churn.
5. Evaluate classification performance.

4.5 Regression Models

Regression is used to estimate MonthlyCharges.

4.5.1 Linear Regression

Linear Regression models the relationship between customer characteristics and monthly charges.

The general equation is:

$Y = b0 + b1X1 + b2X2 + ... + bnXn$

4.5.2 Polynomial Regression

Polynomial Regression captures nonlinear relationships between the input features and monthly charges.

Polynomial features are generated before fitting the regression model.

Model evaluation metrics:

- Mean Absolute Error (MAE).
- Mean Squared Error (MSE).
- Root Mean Squared Error (RMSE).
- R-squared score.

4.6 Customer Segmentation

Customer segmentation identifies groups of customers with similar behavioural characteristics.

4.6.1 K-Means Clustering

K-Means divides customers into K clusters.

The objective is to minimize: $J = \Sigma \Sigma ||xi - \mu j||^2$

where:

- xi represents a customer observation.
- μj represents the centroid of cluster j.
- K represents the number of clusters.

The Elbow Method and Silhouette Score can be used to determine a suitable cluster count.

4.6.2 Hierarchical Clustering

Hierarchical Clustering groups customers based on their similarity.

Agglomerative clustering is used to combine similar customer observations into larger groups.

A dendrogram can be used to visualize the hierarchical structure.

4.7 Model Evaluation

Classification models are evaluated using the following metrics:

Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision

Precision = TP / (TP + FP)

Recall

Recall = TP / (TP + FN)

F1-score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

The confusion matrix is used to analyse correct and incorrect predictions.

For churn prediction, recall for the churn class is particularly important because missing customers who are likely to leave can reduce the effectiveness of retention strategies.

CHAPTER 5: SYSTEM IMPLEMENTATION

5.1 Development Environment

Component	Technology
Programming language	Python
Development environment	Jupyter Notebook / Google Colab
Data manipulation	Pandas
Numerical computation	NumPy
Machine learning	Scikit-learn
Visualization	Matplotlib
Additional visualization	Seaborn
Dataset format	CSV

5.2 Required Libraries

The following Python libraries are used:

- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn

5.3 Implementation Workflow

The implementation is divided into six major modules.

Module 1: Dataset Loading

Import the IBM Telco Customer Churn dataset and inspect the data.

Module 2: Data Cleaning

Handle missing values, duplicates, categorical variables and incorrect data types.

Module 3: Exploratory Analysis

Generate visualizations to understand customer behaviour and churn patterns.

Module 4: Churn Prediction

Implement Logistic Regression, Decision Tree and K-NN.

Module 5: Monthly Charge Estimation

Implement Linear Regression and Polynomial Regression.

Module 6: Customer Segmentation

Apply K-Means and Hierarchical Clustering and analyse the resulting customer groups.

5.4 Model Comparison

The classification algorithms are compared using a common training and testing strategy.

A stratified train-test split is recommended to maintain the churn-class distribution in both datasets.

The dataset can be divided into:

- Training data: 80%.
- Testing data: 20%.

A fixed random state is used to make experiments reproducible.

5.5 Segment Profile Analysis

After clustering, each segment is analysed using the following characteristics:

- Average monthly charges.
- Average customer tenure.
- Contract type distribution.
- Internet service distribution.
- Churn rate.
- Customer payment behaviour.

The profiles help identify customers who require special attention.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Exploratory Data Analysis Results

EDA is used to understand the characteristics of telecom customers.

The analysis focuses on:

1. Customer churn distribution.
2. Monthly charges distribution.
3. Customer tenure distribution.
4. Contract type and churn relationship.
5. Payment method and churn relationship.

The visualizations provide insights into customer behaviour and support the selection of relevant features.

6.2 Classification Performance Comparison

The following table is used to compare the three classification models.

Algorithm	**Accuracy**	**Precision**	**Recall**	**F1-score**
Logistic Regression	To be calculated	To be calculated	To be calculated	To be calculated
Decision Tree	To be calculated	To be calculated	To be calculated	To be calculated
K-NN	To be calculated	To be calculated	To be calculated	To be calculated

The best-performing model should be selected based on the actual test results and the business objective. Accuracy alone should not determine the final model.

6.3 Regression Performance

Linear Regression and Polynomial Regression are compared using error metrics.

Algorithm	MAE	MSE	RMSE	R^2
Linear Regression	To be calculated	To be calculated	To be calculated	To be calculated
Polynomial Regression	To be calculated	To be calculated	To be calculated	To be calculated

A lower error indicates better prediction performance, while a higher R^2 generally indicates that the model explains more variation in monthly charges.

6.4 Customer Segmentation Results

The clustering algorithms identify different customer groups.

The following example profiles are used as interpretation categories. Actual segment labels and characteristics must be assigned based on the generated clustering results.

Segment 1: High-Value Customers

Characteristics:

- Higher monthly charges.
- Longer customer tenure.
- Multiple subscribed services.

Retention strategy:

- Offer loyalty rewards.
- Provide premium customer support.
- Introduce personalized service packages.

Segment 2: Price-Sensitive Customers

Characteristics:

- Relatively lower monthly charges.
- Preference for basic service plans.
- Sensitivity to price changes.

Retention strategy:

- Offer affordable packages.
- Provide discounts on long-term subscriptions.
- Introduce customized billing options.

Segment 3: High-Risk Customers

Characteristics:

- Shorter customer tenure.
- Potentially higher churn probability.
- Possible dissatisfaction with service or contract terms.

Retention strategy:

- Provide personalized retention offers.
- Improve customer support.
- Conduct satisfaction surveys.
- Offer suitable contract options.

These are illustrative business profiles, not predetermined outputs of K-Means or Hierarchical Clustering.

6.5 Business Recommendations

Based on the combined analysis, the following recommendations can be provided:

1. Identify customers with high predicted churn probabilities.
2. Provide targeted retention offers to high-risk customers.
3. Improve technical support and service quality.
4. Introduce loyalty programs for long-term customers.
5. Develop affordable plans for price-sensitive customers.
6. Monitor monthly charges and customer satisfaction.
7. Use customer segmentation to personalize marketing campaigns.

CHAPTER 7: ADVANTAGES AND LIMITATIONS

7.1 Advantages

- Identifies customers who are likely to leave.
- Supports data-driven business decisions.

- Compares multiple machine learning algorithms.
- Estimates monthly customer charges.
- Groups customers based on behavioural similarities.
- Supports targeted retention strategies.
- Uses publicly available data and open-source Python libraries.

7.2 Limitations

- The dataset contains historical information and may not represent current telecom customer behaviour.
- Model performance depends on data quality.
- Customer preferences may change over time.
- Clustering results depend on feature selection and the number of clusters.
- The dataset does not directly contain all factors influencing customer satisfaction.
- Regression estimates are limited to the features and relationships represented in the dataset.

CHAPTER 8: CONCLUSION AND FUTURE ENHANCEMENT

8.1 Conclusion

This project presents a machine learning-based approach for telecom customer churn prediction and customer segmentation.

The IBM Telco Customer Churn dataset is analysed using data preprocessing and exploratory data analysis techniques. Logistic Regression, Decision Tree and K-Nearest Neighbors are implemented to predict customer churn. Their performance is evaluated using training and testing datasets and standard classification metrics.

Linear Regression and Polynomial Regression are applied to estimate monthly customer charges. K-Means and Hierarchical Clustering are used to group customers according to their characteristics and service usage patterns.

The combined analytical approach provides useful information about customer behaviour, churn risk and customer segments. Based on the analysis, telecom companies can develop targeted retention strategies, improve customer relationships and support business growth.

The proposed system demonstrates how machine learning techniques can be integrated to address practical business problems in the telecommunications industry.

8.2 Future Enhancements

The project can be enhanced through the following developments:

1. Implement advanced classification algorithms such as Random Forest, XGBoost and Support Vector Machine.
2. Apply hyperparameter optimization using GridSearchCV and RandomizedSearchCV.
3. Develop a web-based customer churn prediction application.
4. Integrate real-time customer information.
5. Use Explainable AI techniques such as SHAP to understand model predictions.
6. Develop automated customer retention recommendation systems.
7. Incorporate customer feedback and service complaint information.
8. Deploy the model using Flask or Streamlit.

REFERENCES

1. IBM, Telco Customer Churn Dataset, IBM Sample Data and Analytics Resources.
2. Scikit-learn Documentation, Machine Learning in Python. https://scikit-learn.org/
3. McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference.
4. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
5. Han, J., Kamber, M., and Pei, J. (2012). Data Mining: Concepts and Techniques. Morgan Kaufmann.
6. Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media.
7. Jain, A. K. (2010). Data Clustering: 50 Years Beyond K-Means. Pattern Recognition Letters, 31(8), 651–666.
8. IBM, Telco Customer Churn Analytics Dataset Documentation.